The Normal Distribution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Properties of the normal distribution**

State three key properties of a normal distribution's shape.

A normal distribution is: (1) symmetric and bell-shaped, centered at its mean; (2) unimodal (a single peak at the mean); (3) its total area under the curve equals 1, and it never touches the horizontal axis (it extends infinitely in both directions, getting closer and closer to zero).

2 **[Foundational]** • **Target: Properties of the normal distribution**

Explain what the two parameters μ and σ control in a normal distribution's shape.

μ (the mean) controls the location of the center of the distribution (where the peak is). σ (the standard deviation) controls the spread: a larger σ produces a wider, flatter curve, while a smaller σ produces a narrower, taller curve — but both remain symmetric and bell-shaped.

3 **[Foundational]** • **Target: The 68-95-99.7 rule**

Adult male heights are approximately normal with mean 70 inches and standard deviation 3 inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.

The 95% range spans $\mu \pm 2\sigma = 70 \pm 2(3) = 70 \pm 6$, so from 64 to 76 inches.

4 **[Foundational]** • **Target: The 68-95-99.7 rule**

Using the same distribution, find the approximate percentage of adult men taller than 76 inches.

76 inches is $\mu + 2\sigma$. By the 68-95-99.7 rule, 95% of the data lies within 2 standard deviations, leaving 5% split evenly in the two tails: 2.5% above 76 inches and 2.5% below 64 inches.

5 **[Foundational]** • **Target: Standardizing and using the normal distribution**

Using the same height distribution (mean 70, standard deviation 3), find the z-score for a height of 77.5 inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than 77.5 inches.

$z = \frac{77.5 - 70}{3} = \frac{7.5}{3} = 2.5$. $P(X < 77.5) = P(Z < 2.5) \approx 0.9938$, or about 99.38% of men are shorter than 77.5 inches.

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6 [Foundational] • Target: Standardizing and using the normal distribution

Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.

$z_{65} = \frac{65 - 70}{3} \approx -1.67$. $z_{75} = \frac{75 - 70}{3} \approx 1.67$.
$P(65 < X < 75) = P(Z < 1.67) - P(Z < -1.67) \approx 0.9525 - 0.0475 = 0.905$, or about 90.5%.

7 [Foundational] • Target: Visualizing the standard normal distribution

The graph shows the standard normal distribution ($\mu = 0$, $\sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).

a Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.
$P(Z > 1) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$.

b Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$. The standard normal curve is symmetric about $z = 0$. By symmetry, the area to the left of -1 is a mirror image of the area to the right of 1, so $P(Z < -1)$ and $P(Z > 1)$ must be equal.

(a) $P(Z > 1) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$.; (b) The standard normal curve is symmetric about $z = 0$. By symmetry, the area to the left of -1 is a mirror image of the area to the right of 1, so $P(Z < -1)$ and $P(Z > 1)$ must be equal.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Working backward: finding values from percentiles

SAT scores are approximately normal with mean 1050 and standard deviation 150. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.

$x = \mu + z\sigma = 1050 + 1.28(150) = 1050 + 192 = 1242$.

9 [Intermediate] • Target: Working backward: finding values from percentiles

Using the same distribution, find the score that separates the bottom 10% of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.

$x = \mu + z\sigma = 1050 + (-1.28)(150) = 1050 - 192 = 858$.

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10 [Intermediate] • Target: Assessing normality

Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.

A normal probability plot compares the ordered data values to what would be expected if the data were exactly normal. If the plotted points fall close to a straight line, the data is approximately normal. Systematic curvature or bending away from a straight line (such as an S-shape or clear bends at the ends) suggests the data deviates from normality — for example, skewness or heavy tails.

11 [Intermediate] • Target: Applying the normal distribution to real data

Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the z-score for a pregnancy lasting 250 days.

$z = \frac{250 - 266}{16} = \frac{-16}{16} = -1$.

12 [Intermediate] • Target: Applying the normal distribution to real data

Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between 234 and 298 days.

$234 = 266 - 32 = \mu - 2\sigma$ and $298 = 266 + 32 = \mu + 2\sigma$, so this range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of pregnancies fall in this range.

13 [Intermediate] • Target: Combining two independent normal random variables

A vending machine dispenses a soda whose volume is normally distributed with mean 355 mL and standard deviation 4 mL, independently of a snack bag whose weight is normally distributed with mean 50 g and standard deviation 2 g. If T represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of T. (This is a simplified illustrative combination, not a real physical unit.)

For independent normal random variables, means add directly: $\mu_T = 355 + 50 = 405$.

14 [Intermediate] • Target: Combining two independent normal random variables

Battery lifespans are normally distributed with mean 500 hours and standard deviation 40 hours. Find the z-score for a battery lasting 460 hours, and describe its relative position.

$z = \frac{460 - 500}{40} = \frac{-40}{40} = -1$. This battery's lifespan is exactly one standard deviation below the mean.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A company manufactures resistors with resistance values approximately normal, mean 100 ohms and standard deviation 5 ohms. A resistor is considered defective if its resistance is below 90 ohms or above 110 ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.

$90 = 100 - 2(5) = \mu - 2\sigma$ and $110 = 100 + 2(5) = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of resistors fall within this range (not defective), leaving approximately 5% defective (split as roughly 2.5% below 90 ohms and 2.5% above 110 ohms).

16 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using z-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.

Student X: $z = \frac{78 - 70}{8} = \frac{8}{8} = 1$. Student Y: $z = \frac{165 - 150}{12} = \frac{15}{12} = 1.25$. Since $1.25 > 1$, Student Y performed relatively better compared to their own test's distribution of test-takers, even though the raw score gap looks different for each test.

17 [Challenging] • Target: Working backward: finding values from percentiles • Cross-Lesson: 2.01 Sampling Methods & Bias

A sample dataset has $Q_1 = 48$ and $Q_3 = 76$. Using the 1.5 IQR rule: (a) Determine the exact lower and upper outlier fences. (b) If the dataset includes values of 4 and 122, determine whether each is an outlier.

(a) Outlier Fences: Calculate $\text{IQR} = Q_3 - Q_1 = 76 - 48 = 28$. Lower fence: $Q_1 - 1.5(\text{IQR}) = 48 - 1.5(28) = 48 - 42 = \mathbf{6}$. Upper fence: $Q_3 + 1.5(\text{IQR}) = 76 + 1.5(28) = 76 + 42 = \mathbf{118}$. (b) Outlier Status: Since $4 < 6$, 4 is a low outlier. Since $122 > 118$, 122 is a high outlier.

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18 **[Challenging] • Target: Assessing normality • Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A researcher wants to investigate the relationship between the type of pet owned (Dog, Cat, Fish) and the preferred music genre (Rock, Pop, Classical). They collect data and present it in a two-way table. If the table has 3 rows and 3 columns, what is the total number of cells in the table that represent a combination of pet ownership and music genre?

A two-way table is used to display the frequency distribution of two categorical variables. The number of rows corresponds to the categories of one variable, and the number of columns corresponds to the categories of the other variable. The total number of cells in the table is the product of the number of rows and the number of columns. In this case, the number of rows is 3 (representing Dog, Cat, Fish) and the number of columns is 3 (representing Rock, Pop, Classical). Total number of cells = Number of rows $\times$ Number of columns Total number of cells = $3 \times 3 = 9$. Each cell represents a unique combination of pet ownership and music genre, such as 'Dog' and 'Rock', 'Cat' and 'Pop', etc.

19 **[Challenging] • Target: Applying the normal distribution to real data • Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A bag contains marbles of different colors. The probability of drawing a red marble is $P(\text{Red}) = \frac{2}{5}$ and the probability of drawing a blue marble is $P(\text{Blue}) = \frac{1}{3}$. If the draws are independent, what is the probability of drawing a red marble followed by a blue marble?

Since the draws are independent, the probability of two independent events occurring in sequence is the product of their individual probabilities. Let R be the event of drawing a red marble and B be the event of drawing a blue marble. We are given: $P(R) = \frac{2}{5}$ $P(B) = \frac{1}{3}$ The probability of drawing a red marble followed by a blue marble is $P(R \text{ and } B) = P(R) \times P(B)$. $P(R \text{ and } B) = \frac{2}{5} \times \frac{1}{3} = \frac{2 \times 1}{5 \times 3} = \frac{2}{15}$. Therefore, the probability of drawing a red marble followed by a blue marble is $\frac{2}{15}$. Final Answer: $\frac{2}{15}$

20 **[Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A manufacturing company is testing a new quality control process. They have collected data on the number of defects found in batches of a product, categorized by the production shift. The data is summarized in the two-way table below: | Shift | Defects Found | No Defects Found | Total | |---|---|---|---| | Morning | 15 | 85 | 100 | | Afternoon | 25 | 75 | 100 | | Evening | 10 | 90 | 100 | | Total | 50 | 250 | 300 | If a test of independence is to be performed on this data, what is the expected number of defects found in the afternoon shift?

To find the expected number of defects found in the afternoon shift for a test of independence, we use the formula for expected counts: $E_{ij} = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$ In this case, for the cell representing "Afternoon" shift and "Defects Found": - Row Total (Afternoon) = 100 - Column Total (Defects Found) = 50 - Grand Total = 300 Therefore, the expected count is: $E_{\text{Afternoon, Defects Found}} = \frac{100 \times 50}{300} = \frac{5000}{300} = \frac{50}{3} \approx 16.67$

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Properties of the normal distribution

A normal distribution has mean 64 and standard deviation 9. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(46, 82)$ ✓

b. $(37, 91)$

c. $(46, 81)$

d. $(47, 82)$

The middle 95% range spans $\mu \pm 2\sigma = 64 \pm 2(9) = 64 \pm 18$, so from 46 to 82: $(46, 82)$.

22 [Foundational] • Target: The 68-95-99.7 rule

A normal distribution has mean 332 and standard deviation 16. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.

a. $(268, 396)$

b. $(284, 379)$

c. $(285, 380)$

d. $(284, 380)$ ✓

The middle 99.7% range spans $\mu \pm 3\sigma = 332 \pm 3(16) = 332 \pm 48$, so from 284 to 380: $(284, 380)$.

23 [Foundational] • Target: Standardizing and using the normal distribution

A normal distribution has mean 457 and standard deviation 11. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.

a. $(425, 490)$

b. $(424, 490)$ ✓

c. $(424, 489)$

d. $(413, 501)$

The middle 99.7% range spans $\mu \pm 3\sigma = 457 \pm 3(11) = 457 \pm 33$, so from 424 to 490: $(424, 490)$.

24 [Foundational] • Target: Visualizing the standard normal distribution

A normal distribution has mean 50 and standard deviation 14. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

a. $(22, 78)$

b. $(36, 64)$ ✓

c. $(36, 63)$

d. $(37, 64)$

The middle 68% range spans $\mu \pm 1\sigma = 50 \pm 1(14) = 50 \pm 14$, so from 36 to 64: $(36, 64)$.

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25 [Foundational] • Target: Working backward: finding values from percentiles

A normal distribution has mean 138 and standard deviation 40. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(58, 218)$ ✓

b. $(18, 258)$

c. $(58, 217)$

d. $(59, 218)$

The middle 95% range spans $\mu \pm 2\sigma = 138 \pm 2(40) = 138 \pm 80$, so from 58 to 218: $(58, 218)$.

26 [Foundational] • Target: Assessing normality

A normal distribution has mean 88 and standard deviation 18. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.

a. $(70, 106)$ ✓

b. $(70, 105)$

c. $(52, 124)$

d. $(71, 106)$

The middle 68% range spans $\mu \pm 1\sigma = 88 \pm 1(18) = 88 \pm 18$, so from 70 to 106: $(70, 106)$.

27 [Foundational] • Target: Applying the normal distribution to real data

A normal distribution has mean 361 and standard deviation 35. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(291, 430)$

b. $(256, 466)$

c. $(292, 431)$

d. $(291, 431)$ ✓

The middle 95% range spans $\mu \pm 2\sigma = 361 \pm 2(35) = 361 \pm 70$, so from 291 to 431: $(291, 431)$.

28 [Foundational] • Target: Combining two independent normal random variables

A normal distribution has mean $\mu = 101$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 103$, to two decimal places.

a. 69.15% ✓

b. 72.15%

c. 66.15%

d. 30.85%

$z = \frac{103 - 101}{4} = 0.50$. Using the standard normal distribution, $P(Z < 0.50) \approx 0.6915$, or about 69.15%.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Empirical Rule (68-95-99.7)

A normal distribution has mean $\mu = 64$ and standard deviation $\sigma = 10$. Find the approximate percentage of values below $x = 39$, to two decimal places.

a. 0.62% ✓

b. 2.62%

c. 3.62%

d. 99.38%

$z = \frac{39 - 64}{10} = -2.50$. Using the standard normal distribution, $P(Z < -2.50) \approx 0.0062$, or about 0.62%.

30 [Intermediate] • Target: Finding Normal Probabilities

A normal distribution has mean $\mu = 477$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 527$, to two decimal places.

a. 96.38%

b. 0.62%

c. 97.38%

d. 99.38% ✓

$z = \frac{527 - 477}{20} = 2.50$. Using the standard normal distribution, $P(Z < 2.50) \approx 0.9938$, or about 99.38%.

31 [Intermediate] • Target: Inverse Normal Calculations (Finding Values from Probabilities)

A normal distribution has mean $\mu = 174$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 198$, to two decimal places.

a. 6.68%

b. 90.32%

c. 93.32% ✓

d. 96.32%

$z = \frac{198 - 174}{16} = 1.50$. Using the standard normal distribution, $P(Z < 1.50) \approx 0.9332$, or about 93.32%.

32 [Intermediate] • Target: Assessing Normality with Normal Probability Plots

A normal distribution has mean $\mu = 338$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 388$, to two decimal places.

a. 99.38% ✓

b. 96.38%

c. 0.62%

d. 97.38%

$z = \frac{388 - 338}{20} = 2.50$. Using the standard normal distribution, $P(Z < 2.50) \approx 0.9938$, or about 99.38%.

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33 [Intermediate] • Target: Properties of the normal distribution

A normal distribution has mean $\mu = 99$ and standard deviation $\sigma = 2$. Find the approximate percentage of values below $x = 104$, to two decimal places.

a. 99.38% ✓

b. 97.38%

c. 96.38%

d. 0.62%

$z = \frac{104 - 99}{2} = 2.50$. Using the standard normal distribution, $P(Z < 2.50) \approx 0.9938$, or about 99.38%.

34 [Intermediate] • Target: The 68-95-99.7 rule

A normal distribution has mean $\mu = 343$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 335$, to two decimal places.

a. 5.28%

b. 2.28% ✓

c. 4.28%

d. 97.72%

$z = \frac{335 - 343}{4} = -2$. Using the standard normal distribution, $P(Z < -2) \approx 0.0228$, or about 2.28%.

35 [Intermediate] • Target: Standardizing and using the normal distribution

A normal distribution has mean $\mu = 237$ and standard deviation $\sigma = 2$. Find the approximate percentage of values between $x_1 = 233$ and $x_2 = 240$, to two decimal places.

a. 91.04% ✓

b. 87.04%

c. 95.04%

d. 93.32%

$z_1 = \frac{233 - 237}{2} = -2$. $z_2 = \frac{240 - 237}{2} = 1.50$. $P(-2 < Z < 1.50) \approx 0.9104$, or about 91.04%.

36 [Intermediate] • Target: Visualizing the standard normal distribution

A normal distribution has mean $\mu = 123$ and standard deviation $\sigma = 10$. Find the approximate percentage of values between $x_1 = 133$ and $x_2 = 148$, to two decimal places.

a. 11.24%

b. 19.24%

c. 99.38%

d. 15.24% ✓

$z_1 = \frac{133 - 123}{10} = 1$. $z_2 = \frac{148 - 123}{10} = 2.50$. $P(1 < Z < 2.50) \approx 0.1524$, or about 15.24%.

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37 [Intermediate] • Target: Working backward: finding values from percentiles

Scores on an exam are approximately normal with mean $\mu = 837$ and standard deviation $\sigma = 50$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.

a. 909

b. 755

c. 919 ✓

d. 929

$x = \mu + z\sigma = 837 + (1.645)(50) = 919$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Assessing normality • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Scores on an exam are approximately normal with mean $\mu = 295$ and standard deviation $\sigma = 20$. A student scores at the 10th percentile, corresponding to $z \approx -1.28$. Find that student's approximate score.

a. 279

b. 259

c. 321

d. 269 ✓

$x = \mu + z\sigma = 295 + (-1.28)(20) = 269$.

39 [Challenging] • Target: Applying the normal distribution to real data • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A company manufactures parts with a measurement approximately normal, mean 224 and standard deviation 27. A part is considered defective if its measurement is below 170 or above 278 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 32%

b. 2.5%

c. 5% ✓

d. 0.3%

$170 = \mu - 2\sigma$ and $278 = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of parts fall within this range, leaving approximately 5% defective (roughly 2.5% in each tail).

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40 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A company manufactures parts with a measurement approximately normal, mean 84 and standard deviation 15. A part is considered defective if its measurement is below 39 or above 129 (i.e. outside $\mu \pm 3\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 0.3% ✓

b. 5%

c. 0.15%

d. 32%

$39 = \mu - 3\sigma$ and $129 = \mu + 3\sigma$, so the acceptable range spans $\mu \pm 3\sigma$. By the 68-95-99.7 rule, approximately 99.7% of parts fall within this range, leaving approximately 0.3% defective (roughly 0.15% in each tail).

41 [Challenging] • Target: The Empirical Rule (68-95-99.7) • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In a normal distribution with mean $\mu = 100$ and $\sigma = 15$, approximately what percentage of scores fall between 70 and 130?

a. 95% ✓

b. 68%

c. 99.7%

d. 50%

$70 = 100 - 2(15)$ and $130 = 100 + 2(15)$. By the Empirical Rule, approximately 95% of data in a normal distribution falls within 2 standard deviations of the mean.

42 [Challenging] • Target: The Empirical Rule (68-95-99.7) • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

If heights of adult men are $N(70, 3)$ inches, what percentage are taller than 76 inches?

a. 2.5% ✓

b. 5%

c. 16%

d. 0.15%

76 is $70 + 2(3) = \mu + 2\sigma$. Beyond 2 standard deviations is $100\% - 95\% = 5\%$. In the upper tail alone: $\frac{5\%}{2} = 2.5\%$.

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43 [Challenging] • Target: Finding Normal Probabilities • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A machine fills cereal boxes normally with $\mu = 16.2$ oz and $\sigma = 0.1$ oz. What proportion of boxes contain less than the stated label of 16.0 oz?

a. $P(Z < -2.0) \approx 0.0228$ ✓

b. $P(Z < -1.0) \approx 0.1587$

c. $P(Z > 2.0) \approx 0.9772$

d. 0.5000

$z = \frac{16.0 - 16.2}{0.1} = \frac{-0.2}{0.1} = -2.0$. The tail probability is $P(Z < -2.0) \approx 0.0228$ (about 2.28%).

44 [Challenging] • Target: Inverse Normal Calculations (Finding Values from Probabilities) • Cross-Lesson: 1.01 Measures of Center & Spread

In a normal distribution, what are the z-scores corresponding to the first quartile (Q_1) and third quartile (Q_3)?

a. $z \approx -0.67$ **and** $z \approx +0.67$ ✓

b. $z \approx -1.00$ and $z \approx +1.00$

c. $z \approx -0.50$ and $z \approx +0.50$

d. $z \approx -1.96$ and $z \approx +1.96$

Q_1 has 25% of area below it, giving $z \approx -0.674$. Q_3 has 75% below, giving $z \approx +0.674$.

45 [Challenging] • Target: Assessing Normality with Normal Probability Plots • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

What appearance in a Normal Probability (Q-Q) Plot indicates that the sample data comes from a roughly normal population?

a. **The plotted points follow a roughly linear pattern** ✓

b. The plotted points form a bell-shaped curve

c. The plotted points are completely horizontal

d. The plotted points form a curved parabolic shape

If sample data comes from a normal population, the normal quantile plot displays a straight linear trend.

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